

**FH Aachen**

**Department of Electrical Engineering and Information Technology**

# **Exposé**

**Evaluating AI Agent Capabilities in SysML v2 Modeling  
via an MCP Bridge to Magic Systems of Systems Architect**

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1 Exposé

Motivation

[TODO 150 words]

Problem Statement

[TODO 150 words]

Research Question and Hypotheses

**RQ:** Which SysML v2 modeling tasks can an LLM-based agent perform reliably in an industrial modeling environment (MSoSA) when connected through an MCP bridge, and where do its capabilities break down?

- **H1 (Task asymmetry):** Read-only tasks (Abfragen, Validierung) are performed with high reliability, whereas write tasks (Korrektur, Erstellen) and Verifizierung degrade with model size and cross-layer dependency depth.
- **H2 (Tool semantics):** Exposing tool-native capabilities (validation suites, expression evaluation) as MCP tools yields larger gains than a thin CRUD wrapper over the SysML v2 API alone.
- **H3 (Model scale):** Findings obtained on small teaching models (GfSE collection) do not transfer to a large multi-layer model (Apollo 11) without a semantic navigation layer.

Approach

| Use case                     | Example task on Apollo 11                                                        | Ground truth / metric                   |
|------------------------------|----------------------------------------------------------------------------------|-----------------------------------------|
| Abfragen (query)             | “Which technical components satisfy requirement X?”                              | exact set match vs. model query         |
| Validierung (validation)     | “Is package P well-formed and CoSMA-conformant?”                                 | tool validation suite results           |
| Verifizierung (verification) | “Is mass budget requirement R satisfied by the current values?”                  | expression evaluation in tool           |
| Korrektur (correction)       | repair injected faults (dangling ref., wrong type, broken <code>satisfy</code> ) | fault-repair rate, no collateral change |
| Erstellen (creation)         | add missing technical component with ports + satisfy links                       | syntactic validity + expert rubric      |

Table 1: Use-case catalogue (to be refined).

[TODO 150 words prose around the table]

## Evaluation Methodology

*[TODO 150 words]*

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## Bibliography

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model and shows it offers no direct graph traversal, leading to performance bottlenecks on large models; proposes SQLite as an intermediate, traversable representation with SQL as query language (open-source `DLR-FT/sysml-v2-sql`). Method: API analysis plus design and open-source implementation. Technical motivation for the semantic/cache layer of the bridge and for querying inside the tool rather than through the REST API; PDF at `bib/papers/ahlbrecht2025mbsqle.pdf`.

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- [6] DEHART, John K.: Leveraging Large Language Models for Direct Interaction with SysML v2. In: *INCOSE International Symposium 34* (2024), Juli, No. 1, p. 2168–2185. <http://dx.doi.org/10.1002/iis2.13262>. – DOI 10.1002/iis2.13262. – The founding paper of the topic: argues that SysML v2’s English-like textual syntax and the maturation of LLMs make conversational interaction with system models feasible, and that LLMs can replace API expertise as the interface. Method: conceptual analysis plus three OpenAI Assistants-API case studies – (1) modifying a beam model’s length attribute, where the LLM generates and locally executes ad-hoc Python/regex code against the model’s text file (no structured tool/API call); (2) a Flask `/ask` endpoint wrapping the same Assistant for conversational querying, e.g. correctly computing a von Mises stress from boundary conditions given only in prose; (3) a Jupyter-notebook case validating two requirements (max volume, max length) against a block model with a deliberately unconventional calculation, correctly traversing the Definition/Usage relationship and flagging the volume requirement as failed. All three are single-run proofs of concept – no ground truth, no repeated trials, no quantitative metric beyond the reported outcome – and explicitly warns of automation bias and overreliance. Origin of the idea the thesis puts to an empirical test; also the earliest documented instance of the “no structured tool, LLM edits text directly” pattern that motivates this thesis’s no-tool/thin-bridge ablation arms. Full text read 19.09.2026 (INCOSE/Wiley, via FH Aachen library access); PDF at `bib/papers/dehart2024llm.pdf`.

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- [8] QUALIS, Richard A.: Mitigating Hallucinations in SysML v2 Generation Using LLMs and a Tri-Layered Knowledge Graph Reasoning Framework. In: *2025 ACM/IEEE 28th International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, IEEE, Oktober 2025, p. 357–366. – Tri-layer knowledge graph (Layer 1: reusable SysML pattern/annotation KG for nine diagram types; Layer 2: domain-specific KG for sectors like aerospace/automotive; Layer 3: auto-generated system-specific KG parsed from structured requirements) feeding reusable prompt templates to ground SysML v2 generation against hallucinated constructs; satellite-system case study evaluated on a manually curated ground-truth dataset (Average Structure Score, Average Manual Structure Score, Average Semantic Score, Consistency Ratio, prompt-type analysis). Produces structurally valid models reliably; semantic fidelity and output determinism remain limited. Motivates grounding the agent in the *live* model/-tool state rather than a hand-built static KG. Reference point for the *Erstellen* use case alongside [dehn2025nl2sysml](#), [wu2025sysforge](#), [bouamra2025systemp](#). PDF at [bib/papers/qualis2025hallucination.pdf](#).
- [9] QUAST, Vincent ; JACOBS, Georg ; DEHN, Simon ; HÖPFNER, Gregor: Enabling Humans and AI Systems to Retrieve Information from System Architectures in Model-Based Systems Engineering. In: *Systems* 14 (2026), Januar, No. 1, p. 83. <http://dx.doi.org/10.3390/systems14010083>. – DOI 10.3390/systems14010083. – GraphRAG multi-agent system (Supervisor Agent + Graph Query Agent) over a Neo4j knowledge graph parsed from SysML v2 textual notation by a custom Python parser and schema’d by the RFLP methodology; hybrid graph+vector indexing, Cypher

queries, query decomposition, following Peng et al.'s GraphRAG framework. Evaluated on a synthetic battery-EV reference model (4 abstraction levels) with a zero/one-hop vs. multi-hop QA dataset, comparing four LLMs (LLaMA-3.3-70B, Gemini-2.0-flash-lite, Gemini-2.0-flash, Gemini-2.5-flash-preview); best model reaches **93 % overall accuracy, 90 % on multi-hop questions**. Authors' own stated limitations: metamodel-specific parser, synthetic (not industrial-scale) model, no user studies, and explicitly "*lack of standardized benchmarks*" for QA-over-MBSE – an independent second data point for this thesis's benchmark-gap argument (alongside bouamra2025systemp). Primary reference for the *Abfragen* use case and its accuracy metric; the thesis queries the live tool/model state directly rather than a derived, pre-built graph. PDF at <bib/papers/quast2026graphrag.pdf>.

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three-arm ablation (no tool / thin bridge / semantic bridge).

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Satisfaction Problem and solves it via constraint propagation – explicitly static, explicitly excludes dynamic/behavioral analysis (“restrict constraint propagation to... the static structure... excluding dynamic behaviors”). Uses “valid/validation” only informally for constraint satisfaction. Audited 19.09.2026, verdict: Verification. PDF at [bib/papers/ratzke2025constraints.pdf](#).

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